

When Metrics Disagree: A Meta-Analysis of Knowledge-Graph-Completion Model Benchmarking

(Supplementary Material)

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This supplementary document provides the complete baseline and ablation results for the multi-criteria decision-making (MCDM) meta-evaluation of knowledge graph completion (KGC) models. Unlike the main manuscript, which reports results averaged across the combined leave-one-model-out (LOMO) and leave-one-group-out (LOGO) removal scheme, this supplement presents three distinct configurations: (1) **Baseline/Full-Set**: analysis with all models included (no removal), establishing reference rankings and correlation structures; (2) **LOMO-only**: overall scores computed after sequentially removing individual KGC models; and (3) **LOGO-only**: overall scores computed after removing entire architectural families (translational/geometric, semantic-matching, neural-CNN, GNN-based). All results cover the tail prediction task $(h, r, ?)$ with 20 models across five benchmarks (FB15K, WN18, FB15K-237, WN18RR, YAGO3-10) and the relation prediction task $(h, ?, t)$ with 12 models across six benchmarks (adding NELL995 and DDB14), evaluated using seven MCDM aggregators (EDAS, TOPSIS, VIKOR, MOORA, WASPAS, Borda Count, Z-Score) across five reliability dimensions. The following sections report detailed aggregation scores, model rankings, consistency matrices, stability distributions, independence tests, robustness under noise injection, generalizability validation, and Pareto optimality summaries for each configuration.

1 Base Results and Analysis

This section reports results for two KGC prediction tasks: relation prediction $(h, ?, t)$ and tail prediction $(h, r, ?)$. The analysis begins with the tail prediction task, where seven MCDM aggregators are evaluated across five dimensions: consistency, stability, metric independence, robustness, and generalizability.

To ensure fair comparison across heterogeneous evaluation criteria, all metrics are normalized so that

higher values indicate better performance and are bounded by 1.0. Specifically, MRR is retained in its native $[0, 1]$ scale; Hits@1 and Hits@10 are scaled by dividing each model’s value by the column maximum; and MR, being a non-beneficial metric, is transformed using $MR_{\text{norm}} = \min(\text{MR})/\text{MR}$. This approach ensures that the best-performing model reaches 1.0 while preserving the relative ordering of models and avoiding zero values that could affect multiplicative aggregators.

1.1 Tail Prediction Analysis

Table 1 reports the base tail prediction results across five benchmark datasets. A clear pattern emerges: model performance is highly fragmented across metrics and datasets. No single KGC architecture dominates all evaluation criteria, and strong performance on one benchmark does not consistently transfer to others. For example, DualE achieves the best MR on FB15k, but this advantage is not consistent across other datasets. In contrast, models such as ComplEx, RotatE, TuckER, and Hake show varying strengths depending on the metric considered. This confirms that comparisons based on individual metrics can lead to inconsistent conclusions. The role of MCDM is therefore not to replace the original metrics, but to combine these fragmented signals into a more stable and interpretable ranking.

The corresponding aggregation scores and model rankings are reported in Tables 2 and 3. These results show that the choice of aggregator influences the final ranking. While the top models are generally similar, their ordering changes across EDAS, TOPSIS, VIKOR, Borda, Z-Score, MOORA, and WASPAS. The next part of this section evaluates these aggregators using the five criteria to identify the most reliable method for tail prediction.

Consistency Test: To evaluate the consistency

Table 1: Link prediction results (head/tail entity prediction) on FB15k, WN18, FB15k-237, WN18RR, and YAGO3-10. Results reported in this table are published in (1; 2; 3; 4; 5).

2*Models	FB15k				WN18				FB15k-237				WN18RR				YAGO3-10			
	MR ↓	MRR ↑	H@1 ↑	H@10 ↑	MR ↓	MRR ↑	H@1 ↑	H@10 ↑	MR ↓	MRR ↑	H@1 ↑	H@10 ↑	MR ↓	MRR ↑	H@1 ↑	H@10 ↑	MR ↓	MRR ↑	H@1 ↑	H@10 ↑
Translational / Geometric Models																				
TransE	45	0.628	0.494	0.847	279	0.646	0.406	0.949	209	0.310	0.217	0.497	3936	0.206	0.028	0.495	1187	0.501	0.406	0.674
STransE	69	0.543	0.398	0.796	208	0.656	0.431	0.935	357	0.315	0.225	0.496	5172	0.226	0.101	0.422	5797	0.049	0.033	0.074
CrossE	136	0.702	0.601	0.862	441	0.834	0.733	0.950	227	0.298	0.212	0.471	5212	0.405	0.381	0.450	3839	0.446	0.331	0.654
TorusE	143	0.746	0.689	0.840	525	0.947	0.943	0.954	211	0.281	0.196	0.447	4873	0.463	0.427	0.534	1945	0.342	0.274	0.474
RotatE	42	0.791	0.739	0.881	274	0.949	0.943	0.960	178	0.336	0.238	0.531	3318	0.475	0.426	0.573	1827	0.498	0.405	0.671
HakE	128	0.714	0.639	0.834	304	0.934	0.919	0.957	160	0.346	0.250	0.542	2250	0.497	0.452	0.582	1130	0.545	0.462	0.694
DualE	21	0.813	0.766	0.896	156	0.952	0.946	0.962	91	0.365	0.268	0.559	2270	0.492	0.444	0.584	2068	0.280	0.096	0.591
Semantic Matching Models																				
DistMult	173	0.784	0.736	0.863	675	0.824	0.726	0.946	199	0.313	0.224	0.490	5913	0.433	0.397	0.502	1107	0.501	0.413	0.661
ComplEx	34	0.848	0.816	0.905	3623	0.949	0.945	0.955	202	0.349	0.257	0.530	4907	0.458	0.426	0.521	1112	0.576	0.505	0.704
ANALOGY	126	0.726	0.656	0.837	808	0.934	0.926	0.944	476	0.202	0.126	0.354	9266	0.366	0.358	0.380	2423	0.283	0.192	0.457
SimplE	138	0.726	0.661	0.836	759	0.938	0.933	0.946	651	0.179	0.100	0.344	8764	0.398	0.383	0.427	2849	0.453	0.358	0.632
HolE	211	0.800	0.759	0.868	650	0.938	0.931	0.949	186	0.303	0.214	0.476	8401	0.432	0.403	0.488	6489	0.502	0.418	0.652
TuckER	39	0.788	0.729	0.889	510	0.951	0.946	0.958	162	0.352	0.259	0.536	6239	0.459	0.430	0.514	2417	0.544	0.466	0.681
Neural / CNN-based Models																				
ConvE	51	0.688	0.595	0.849	413	0.945	0.939	0.957	281	0.305	0.219	0.476	4944	0.427	0.390	0.508	2429	0.488	0.399	0.658
ConvKB	109	0.567	0.462	0.762	516	0.824	0.743	0.945	257	0.396	0.300	0.517	2554	0.248	0.150	0.525	2626	0.419	0.323	0.586
ConvR	70	0.773	0.723	0.870	471	0.950	0.946	0.959	251	0.346	0.256	0.526	5646	0.467	0.437	0.527	2582	0.527	0.446	0.673
RSN	51	0.777	0.706	0.886	346	0.928	0.912	0.951	248	0.280	0.198	0.444	4210	0.395	0.346	0.483	1339	0.511	0.427	0.664
GNN-based Models																				
R-GCN	68	0.651	0.541	0.825	510	0.819	0.697	0.964	600	0.164	0.100	0.300	6700	0.123	0.080	0.200	15000	0.120	0.060	0.211
CompGCN	69	0.419	0.270	0.498	539	0.676	0.518	0.921	197	0.355	0.264	0.535	3533	0.479	0.443	0.546	1591	0.227	0.126	0.442
NodePiece	420	0.148	0.078	0.158	633	0.435	0.263	0.647	230	0.256	0.190	0.420	5000	0.403	0.400	0.515	3800	0.247	0.200	0.488

Table 2: Aggregation scores for knowledge graph embedding models (ZS: Z-Score, ED: EDAS, MO: MOORA, BS: Borda Score, TP: TOPSIS, WA: WASPAS, VQ: VIKOR).

Model	ED	TP	VQ	BS	ZS	MO	WA	MR	MRR	H1	H10
DualE	0.9397	0.7670	0.1633	1.0000	1.0000	0.2861	0.8403	0.9053	0.5804	0.5040	0.7184
HakE	0.8548	0.6197	0.1446	0.8920	0.8968	0.2609	0.8083	0.6451	0.6072	0.5444	0.7218
ComplEx	0.8269	0.5981	0.1031	0.9379	0.8641	0.2543	0.5132	0.5130	0.6360	0.5898	0.7230
RotatE	0.7997	0.6503	1.0000	0.9246	0.8608	0.2551	0.7947	0.5729	0.6098	0.5502	0.7232
TuckER	0.7459	0.5970	0.1412	0.9275	0.8149	0.2449	0.7476	0.4449	0.6188	0.5660	0.7156
ConvR	0.6842	0.5415	0.1534	0.8328	0.7593	0.2323	0.7122	0.3642	0.6126	0.5616	0.7110
RSN	0.6673	0.5792	0.1707	0.6494	0.7007	0.2337	0.6947	0.5181	0.5782	0.5178	0.6856
DistMult	0.6309	0.5211	0.1004	0.6317	0.6758	0.2243	0.6529	0.4381	0.5710	0.4992	0.6924
ConvE	0.6007	0.5374	0.1386	0.6257	0.6663	0.2230	0.6682	0.4048	0.5706	0.5084	0.6896
HolE	0.5608	0.4808	0.0885	0.6080	0.6323	0.2108	0.5861	0.2534	0.5950	0.5450	0.6866
TorusE	0.5349	0.4860	0.0983	0.6109	0.5996	0.2094	0.6150	0.3812	0.5558	0.5058	0.6498
TransE	0.4915	0.5446	0.1165	0.5799	0.5438	0.2132	0.5863	0.5931	0.4582	0.3102	0.6924
CrossE	0.4774	0.4729	0.0954	0.4882	0.5512	0.2032	0.5875	0.3258	0.5370	0.4516	0.6774
ConvKB	0.4042	0.4394	0.0780	0.5163	0.5377	0.1954	0.3840	0.4303	0.4908	0.3356	0.6670
SimplE	0.3933	0.4272	0.0714	0.3876	0.4034	0.1834	0.2853	0.2286	0.5388	0.4870	0.6370
CompGCN	0.3867	0.4676	0.1049	0.5814	0.4569	0.1943	0.5386	0.4777	0.4312	0.3242	0.5884
ANALOGY	0.3134	0.3957	0.0681	0.3402	0.3380	0.1721	0.3023	0.2501	0.5022	0.4516	0.5944
STransE	0.1313	0.3802	0.0624	0.3358	0.2214	0.1557	0.2102	0.3870	0.3578	0.2376	0.5446
NodePiece	0.0321	0.3389	0.0567	0.2840	0.0010	0.1404	0.1550	0.2867	0.2978	0.2262	0.4456
R-GCN	0.0071	0.2938	0.0564	0.2722	0.0258	0.1289	0.1524	0.2352	0.3754	0.2956	0.5000

Table 3: Ranking of knowledge graph completion models based on MCDM aggregator methods (1 = best).

Model	ED	TP	VQ	BS	ZS	MO	WA	MR	MRR	H1	H10
DualE	1	1	3	1	1	1	1	1	7	10	4
HakE	2	3	5	5	2	2	2	2	5	6	3
ComplEx	3	4	10	2	3	4	14	6	1	1	2
RotatE	4	2	1	4	4	3	3	4	4	4	1
TuckER	5	5	6	3	5	5	4	8	2	2	5
ConvR	6	8	4	6	6	7	5	14	3	3	6
RSN	7	6	2	7	7	6	6	5	8	7	11
DistMult	8	10	11	8	8	8	8	9	9	11	7
ConvE	9	9	7	9	9	9	7	11	10	8	9
HolE	10	12	14	11	10	11	12	17	6	5	10
TorusE	11	11	12	10	11	12	9	13	11	9	14
TransE	12	7	8	13	13	10	11	3	16	17	7
CrossE	13	13	13	15	12	13	10	15	13	13	12
ConvKB	14	15	15	14	14	14	15	10	15	15	13
Simple	15	16	16	16	16	16	17	20	12	12	15
CompGCN	16	14	9	12	15	15	13	7	17	16	17
ANALOGY	17	17	17	17	17	17	16	18	14	13	16
STransE	18	18	18	18	18	18	18	12	19	19	18
NodePiece	19	19	19	19	20	19	19	16	20	20	20
R-GCN	20	20	20	20	19	20	20	19	18	18	19

of each MCDM aggregator, three correlation measures, Kendall’s τ , Pearson’s r , and Spearman’s ρ , are computed between each aggregator’s ranking-assigned values and the base performance metrics (MRR, Hits@1, Hits@10, MR). These correlations are averaged across all benchmarks to assess how well each aggregator preserves the information contained in the original metrics. Higher values indicate stronger agreement and better consistency.

The consistency score for each aggregator is defined as the mean of these three correlations. To facilitate comparison, we normalize the mean scores across all methods. In addition, we quantify stability by computing the standard deviation of the three correlation coefficients, which serves as an error indicator of internal agreement among the correlation measures.

Figure 1 (a) presents the normalized consistency scores together with their corresponding standard-deviation error bars. Z-Score achieves the highest consistency while exhibiting minimal variation, indicating strong and stable maintenance of the base performance ordering. Most aggregators maintain high consistency. However, VIKOR shows a noticeable drop in mean consistency and an increase in variability, suggesting greater sensitivity to the base metric characteristics.

Figure 1 (b) shows the correlation between each MCDM aggregator and each base evaluation metric. Each cell represents the average correlation across all benchmarks. A consistent pattern is observed: correlations with MR are generally lower across all aggregators. This occurs because MR is highly sensitive to large rank differences, which reduces its agreement with other metrics. In contrast, MRR and Hits@ k show stronger and more consistent alignment with the aggregated results. Overall, the results show that the Z-score is the most consistent and reliable aggre-

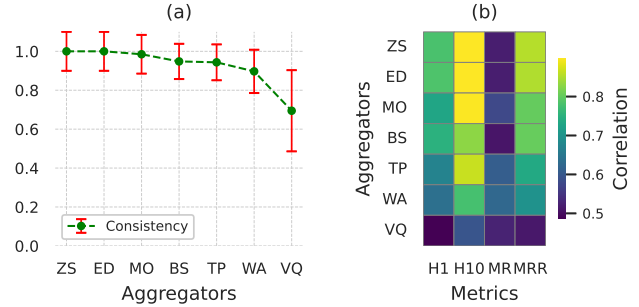


Figure 1: (a) Consistency test of MCDM aggregators. (b) Correlation between aggregators and base metrics.

gator.

For tail prediction, EDAS, TOPSIS, and MOORA form a high-consensus cluster ($\tau \geq 0.911$) due to their shared reference-point normalization, while Borda and Z-Score also align closely. VIKOR and WASPAS exhibit lower agreement, reflecting their distinct formulations.

Stability Test: To assess cross-dataset stability, model rankings are first computed independently for each dataset. For each model, the variance of its ranks across datasets is calculated. Since higher variance indicates greater instability, this value is normalized and inverted so that higher scores indicate more stable rankings. The final stability score is obtained by averaging across all models and tasks for each aggregator.

Figure 2 presents these results using boxplots showing interquartile ranges and whisker spreads. TOPSIS exhibits the highest stability, followed by MOORA and EDAS, indicating stronger cross-dataset consistency, while Borda shows the lowest stability (see Table 5). These results suggest that scoring-based aggregators tend to produce more stable rankings across heterogeneous benchmarks.

Independency Test: To assess aggregator robustness under partial information loss, we employ a leave-one-metric-out analysis test. Each base metric is systematically omitted, and the resulting aggregator rankings are compared against the baseline computed using the complete metric set. This evaluates whether each method maintains ranking stability when evaluation information is incomplete.

For each aggregator, we compute three correlation coefficients between the reduced-metric and full-metric outputs. The overall independency score is the mean of these correlations. Scores are normalized by the maximum value across all methods for easier in-

Table 4: Correlation Matrices for MCDM Methods with Key Metrics, $(h, ?, t)$

MCDM	Kendall τ ($\uparrow$)				Pearson r ($\uparrow$)				Spearman ρ ($\uparrow$)				Overall
	H1	H10	MR	MRR	H1	H10	MR	MRR	H1	H10	MR	MRR	Corr.
EDAS	0.6596	0.8179	0.3684	0.7368	0.8442	0.9280	0.6112	0.9058	0.8409	0.9462	0.5729	0.9038	0.7613
Z-Score	0.6702	0.8074	0.3579	0.7474	0.8222	0.9466	0.6247	0.9029	0.8432	0.9379	0.5714	0.9038	0.7611
MOORA	0.6069	0.8285	0.4211	0.6842	0.7802	0.9080	0.6650	0.8538	0.7943	0.9507	0.6436	0.8617	0.7498
Borda	0.6596	0.7546	0.3684	0.7158	0.7436	0.8260	0.5792	0.7995	0.8424	0.9033	0.5789	0.8902	0.7218
TOPSIS	0.5752	0.8074	0.4526	0.6316	0.6837	0.8404	0.6847	0.7573	0.7507	0.9395	0.6842	0.8120	0.7183
WASPAS	0.5013	0.6596	0.5053	0.5579	0.7314	0.8640	0.6570	0.8012	0.6769	0.8041	0.7083	0.7278	0.6829
VIKOR	0.4802	0.6491	0.5263	0.5158	0.3045	0.3312	0.3004	0.3176	0.6709	0.8011	0.7459	0.7023	0.5288

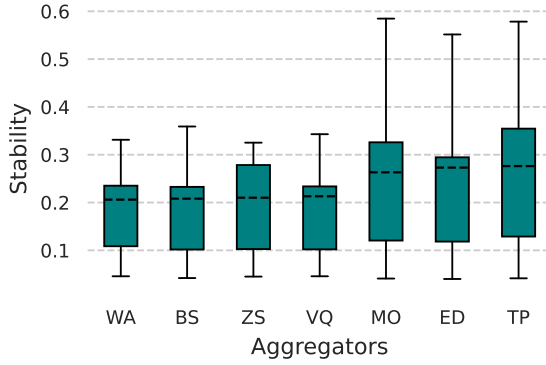


Figure 2: Distribution of stability across different aggregators.

Table 5: Variance of rankings for each knowledge graph embedding model across the five benchmark datasets, computed separately for each MCDM method. Variance is inverted so that higher variance indicates higher ranking stability (more consistent performance) across datasets.

Model	VQ	WA	BC	ZS	ED	MO	TP
ConvR	3.100	3.160	2.400	2.800	2.480	2.880	2.280
RotatE	3.380	3.440	4.200	4.500	3.500	3.660	3.700
CrossE	5.960	7.060	6.240	6.460	2.280	2.300	2.140
TuckER	9.240	6.580	7.240	6.460	3.880	4.360	5.200
ConvE	9.340	6.880	8.640	7.420	5.360	5.320	5.640
ComplEx	9.060	7.920	9.340	11.020	8.240	7.120	6.180
DistMult	13.740	12.280	11.680	12.940	9.360	10.040	9.100
SimplE	7.920	9.840	9.460	9.840	13.980	14.860	14.040
NodePiece	11.940	15.740	13.680	13.600	10.880	10.880	11.200
ANALOGY	10.380	13.960	14.300	14.300	12.960	13.740	14.100
HolE	18.700	17.820	17.340	15.740	12.420	13.180	12.540
RSN	17.280	15.580	14.180	14.380	18.400	16.680	16.160
TransE	15.660	15.000	15.400	15.500	17.980	17.700	17.100
HakE	21.460	17.000	16.980	17.600	19.700	17.920	16.200
STransE	20.880	21.700	22.640	20.940	15.480	15.100	14.240
TorusE	20.440	20.560	20.460	19.320	15.500	18.100	19.260
R-GCN	23.340	26.200	25.000	25.420	17.540	16.620	16.440
DualE	25.680	22.820	23.620	23.660	29.900	27.880	26.880
ConvKB	29.180	34.060	31.020	31.500	31.720	32.720	32.240
CompGCN	50.880	47.420	46.720	46.740	53.340	52.140	51.540
Mean Variance	0.9042	0.9113	0.9240	0.9252	0.9714	0.9768	1.0000

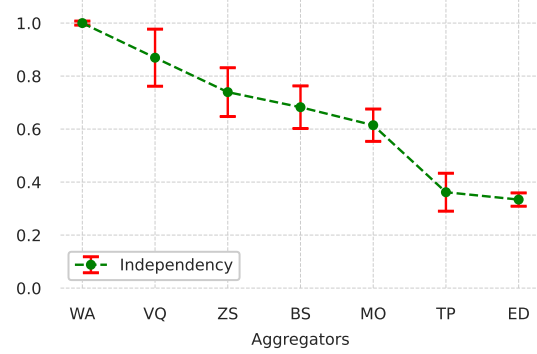


Figure 3: Independency analysis of MCDM aggregators under metric removal.

terpretation. To quantify stability, the analysis also reports the standard deviation across the three correlation measures as error bars, reflecting the internal consistency of agreement under metric removal.

Figure 3 presents the normalized correlation with error bars representing ± 1 standard deviation, sorted from highest to lowest correlation. WASPAS and VIKOR achieve the highest independency, indicating strong resistance to the removal of any single metric. The Z-score achieves near-perfect normalized correlation with the smallest variation. In contrast, TOPSIS and EDAS exhibit substantially lower scores and higher variability, revealing greater sensitivity to the incomplete evaluation metric. These results position WA and VQ, followed by Z-score, as the most robust aggregators for scenarios involving incomplete or fluctuating evaluation.

Table 6 reports the detailed metric-removal results. Kendall’s τ and Spearman’s ρ measure global ranking preservation, average absolute rank displacement ($|\Delta \text{Rank}|$) measures local positional changes, and Top-5 Jaccard similarity evaluates whether the highest-performing models remain stable. This further verifies that Z-score and Borda provide stronger overall balance after normalization.

Robustness Test: This test evaluates aggregator

Table 6: Detailed independency analysis of MCDM aggregation methods under metric removal. For Kendall’s τ , Spearman’s ρ , and Top-5 Jaccard, higher is better; for $|\Delta\text{Rank}|$, lower is better.

Removed	Metric	Borda	Z-score	TOPSIS	VIKOR	EDAS	MOORA	WASPAS
Kendall τ	Without MR	0.495	0.579	0.227	0.663	0.126	0.505	1.000
	Without MRR	0.635	0.674	0.164	0.684	0.189	0.474	1.000
	Without H@1	0.638	0.579	0.575	0.853	0.600	0.589	1.000
	Without H@10	0.505	0.632	0.164	0.716	0.274	0.547	1.000
Spearman ρ	Without MR	0.644	0.723	0.302	0.841	0.093	0.644	1.000
	Without MRR	0.780	0.791	0.180	0.836	0.266	0.586	1.000
	Without H@1	0.776	0.735	0.743	0.941	0.698	0.708	1.000
	Without H@10	0.657	0.761	0.147	0.880	0.362	0.588	1.000
$ \Delta\text{Rank} $	Without MR	1.50	1.60	4.85	3.50	7.60	2.80	2.90
	Without MRR	0.80	0.70	0.65	0.50	0.00	1.60	0.70
	Without H@1	0.60	0.50	1.25	1.40	0.00	1.70	1.50
	Without H@10	1.40	1.20	0.55	0.90	0.00	1.10	0.80
Top-5 Jaccard	Without MR	0.667	0.667	0.111	0.667	0.000	0.429	0.429
	Without MRR	1.000	1.000	1.000	1.000	1.000	0.667	0.667
	Without H@1	1.000	1.000	1.000	0.667	1.000	0.667	0.667
	Without H@10	1.000	0.667	1.000	1.000	1.000	1.000	1.000

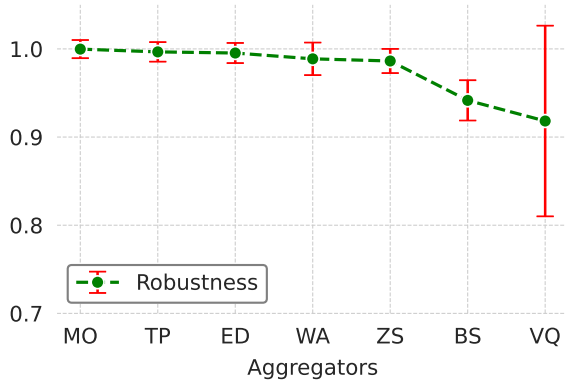


Figure 4: Robustness analysis under noise injection.

stability under measurement uncertainty by injecting noise into metric scores. Each noise level (5%, 10%, 20%, and 30%), repeating 1,000 times per level, to compute average rank displacement and correlation with baseline rankings.

As shown in Figure 4, MOORA, TOPSIS, EDAS, WASPAS, and Z-Score maintain near-perfect normalized correlation with minimal variance, demonstrating strong robustness to metric noise injection across all levels. VIKOR shows the lowest correlation and the widest error bounds, indicating higher sensitivity to noise injection. The tight error margins for top-performing methods suggest that rank- and distance-based aggregators preserve ranking structure reliably even under substantial metric variation, making them suitable for deployment in noisy or uncertain evaluation environments.

For clarity, Figure 5 presents detailed results across noise levels, with aggregators sorted by performance at 30% noise. The *left* panel shows the mean absolute rank displacement ($|\Delta\text{Rank}|$), which is inverted so

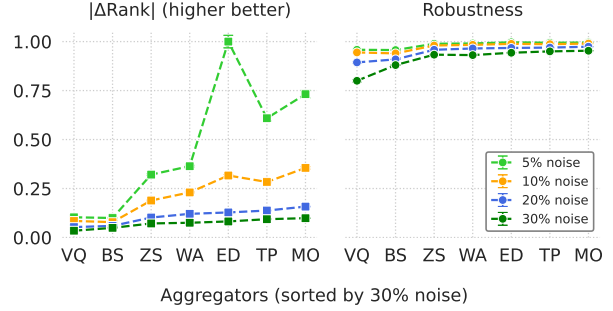


Figure 5: Robustness assessment with noise injection (1,000 iterations per level). *Left*: Mean absolute rank displacement ($|\Delta\text{Rank}|$, higher = more robust). *Right*: average correlation with baseline ranking (higher = more robust).

that higher values indicate greater stability. VIKOR exhibits the highest sensitivity due to its distance-to-ideal formulation, while MOORA and TOPSIS demonstrate superior robustness, maintaining minimal displacement even at 30% noise, as their normalization procedures inherently dampen noise effects. The *right* panel shows Kendall’s correlation with baseline rankings, where all aggregators maintain a strong correlation up to 20% noise. MOORA and TOPSIS preserve the strongest correlation at 30% noise, whereas VIKOR shows the weakest. Overall, MOORA emerges as the most robust aggregator for uncertain KGC evaluations, while VIKOR’s compromise programming proves unsuitable for high-noise environments.

Generalizability Test: This test evaluates cross-dataset transferability using leave-one-dataset-out validation. For each eliminated benchmark, the aggregators are assessed according to how well rankings derived from the remaining datasets transfer to the unseen dataset. The aim is to examine whether each MCDM aggregation method produces rankings that remain reliable across changing evaluation contexts.

For each validation step, one dataset is treated as unseen, while the remaining datasets are used to predict the reference ranking for each aggregator. This transferred ranking is then compared with the ranking obtained directly on the eliminated dataset. The relationships are measured using correlation, which is averaged to obtain an overall generalizability and normalized with $value/Max$.

Figure 6 presents the generalizability with error bars representing ± 1 standard deviation, sorted from highest to lowest score. The results reveal that Z-scores achieve the highest generalizability with min-

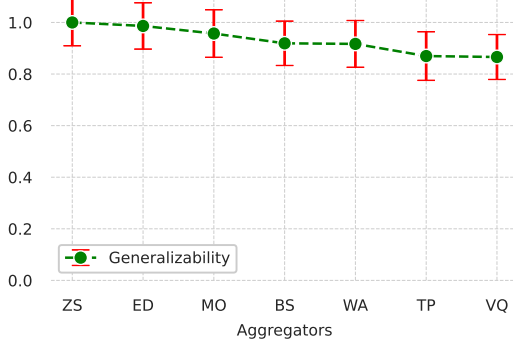


Figure 6: Generalizability analysis of MCDM aggregators under leave-one-dataset-out validation.

Table 7: Generalizability results for tail prediction using leave-one-dataset-out validation. Higher values indicate better cross-dataset transferability.

Aggregator	Top-1	Top-3	τ	Pearson	Spearman	95% CI
Mean Z-Score	0.100	0.250	0.256	0.340	0.340	[0.128, 0.496]
EDAS	0.050	0.150	0.249	0.337	0.337	[0.125, 0.491]
MOORA	0.100	0.250	0.248	0.323	0.323	[0.110, 0.486]
Borda Score	0.050	0.200	0.226	0.317	0.317	[0.111, 0.462]
WASPAS	0.050	0.250	0.238	0.310	0.310	[0.101, 0.471]
TOPSIS	0.050	0.250	0.222	0.296	0.296	[0.079, 0.463]
VIKOR	0.050	0.100	0.220	0.295	0.295	[0.093, 0.447]

imal variation. In contrast, TOPSIS and VIKOR yield the lowest scores, with the highest variation. This indicates that reference-distance-based methods are particularly sensitive to changes in the evaluation context. These results suggest that ZS-based methods offer the most reliable generalization behavior.

Table 7 reports the detailed leave-one-dataset-out results. In each fold, Top-1 accuracy is computed by checking whether the model ranked first by the transferred ranking is also the best model on the eliminated dataset. Similarly, Top-3 accuracy checks whether the best model on the eliminated dataset appears among the top three models in the transferred ranking. The table also reports correlation to measure full-ranking agreement, while the last column provides the corresponding 95% confidence interval. The overall findings of this test show that the Z-score and EDAS are more reliable when rankings learnt from observed datasets are transferred to unseen benchmarks.

1.1.1 Pareto Analysis

To identify the most suitable MCDM aggregator for KGC evaluation, we conduct a Pareto optimality analysis across all five test dimensions. Since no single aggregator dominates all criteria, we employ three complementary visualization and statisti-

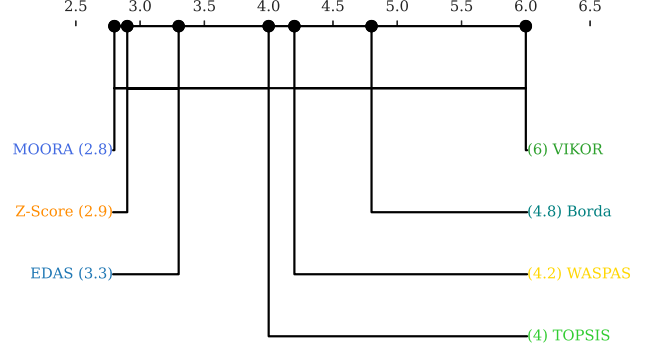


Figure 7: Critical difference (CD) diagram showing average ranks of MCDM aggregators across five evaluation dimensions. Methods connected by horizontal lines are not significantly different (Nemenyi post-hoc test, $\alpha = 0.05$). Z-Score and MOORA achieve the top ranks performance.

cal techniques to determine which methods offer the best trade-offs.

Figure 7 presents a critical difference (CD) diagram based on Friedman ranking tests. The Friedman test determines whether observed differences are statistically significant ($p < 0.05$), followed by Nemenyi post-hoc analysis to compute the critical difference threshold. Z-Score and MOORA achieve the best average ranks, followed by EDAS, while VIKOR performs the worst.

Figure 8 shows normalized performance scores (0–1 scale, higher is better) across all five tests. The radar plot further verified that no aggregator dominates all dimensions, but Z-Score performs most consistently, with consistently high scores across all criteria.

Figure 9 presents the overall performance score for each aggregator across all five tests: $\text{Overall}_i = \frac{1}{5} \sum_{j=1}^5 s_{ij}$. Z-Score achieves the highest overall performance, followed by Borda, MOORA, WASPAS, and EDAS, while VIKOR performs the worst.

1.1.2 KGC method removal method analysis

Additionally, the overall score is computed after one KGC method removal analysis, followed by all five dimensional tests, and the output is reported in Figure 10.

Similarly, the group KGC method removal analysis was performed under each of the five dimensional tests, and the overall score is shown in Figure 11.

Based on the Pareto optimality and KGC method removal analysis, **Z-Score** is selected as the optimal aggregator for holistic KGC evaluation. Table 8 presents the top three KGC methods for the

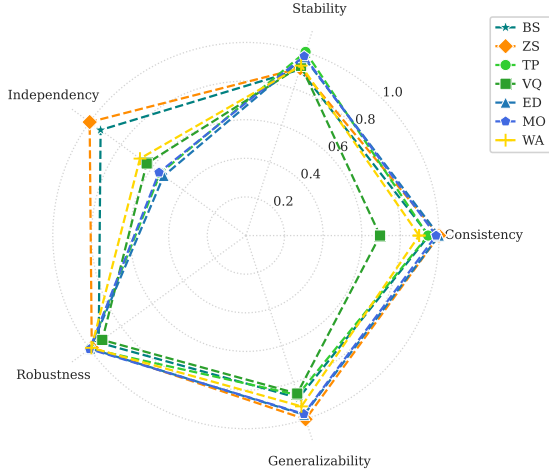


Figure 8: A radar plot showing normalized performance (0–1 scale, higher is better) across five evaluation dimensions. The Z-score demonstrates the most balanced performance with strong scores across consistency, stability, and generalisability.

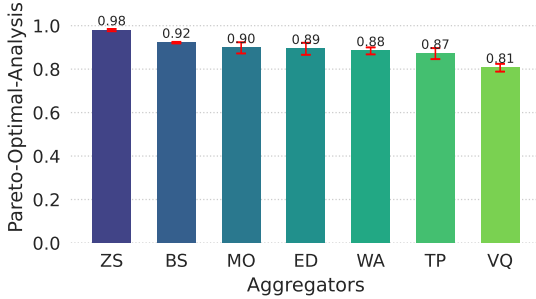


Figure 9: Overall performance scores for each MCDM aggregator. Z-Score achieves the highest score, indicating balanced performance across all tests.

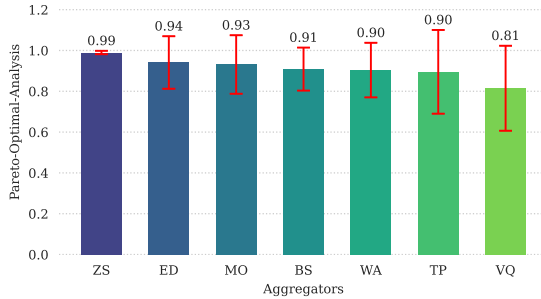


Figure 10: One KGC method removal overall performance.

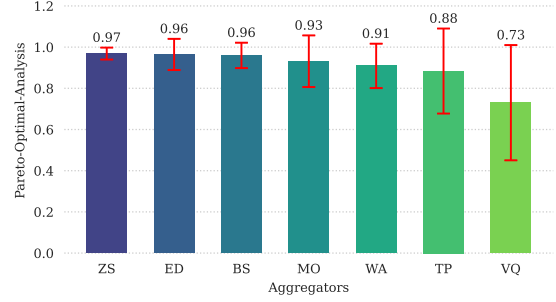


Figure 11: Group KGC method removal overall performance.

Table 8: Top-3 KGC models ranked by the Z-Score aggregator (1 = best).

Model	Z-Score		
	DualE	HakE	ComplEx
Rank	1	2	3

tail prediction task as ranked exclusively by Z-Score. DualE achieves the highest Z-Score aggregate ranking (rank 1), indicating it provides the most balanced performance across all metrics (MR, MRR, Hits@1, and Hits@10) on the tail prediction task.

1.2 Relation Prediction

We apply the same meta-evaluation pipeline to the relation prediction task $(h, ?, t)$. This task uses MRR, Hits@1, and Hits@3 across six benchmark datasets: FB15K, FB15K-237, WN18, WN18RR, NELL995, and DDB14. Base results are reported in Table 15, while MCDM aggregation scores and induced rankings appear in Tables 16 and 17.

Consistency Test: To compute consistency for relation prediction, we follow the same procedure described in the tail prediction consistency test. Figure 19 presents the consistency analysis for relation prediction.

Figure 19 (a) shows normalized consistency scores with standard-deviation error bars: all aggregators maintain near-perfect consistency (≈ 1.0) with minimal variation, indicating strong preservation of base metric rankings. MOORA, Z-Score, EDAS, TOPSIS, WASPAS, and Borda Score exhibit virtually identical performance, while VIKOR shows a marginal decline with slightly elevated variance. Figure 19 (b) depicts pairwise agreement between aggregators and base metrics, revealing a strong correlation between Z-Score and MOORA, whereas VIKOR produces the most distinct ranking behavior. Overall, these results

Table 9: Transductive performance comparison on all six datasets (results are taken from Flow-Modulated Scoring for Semantic-Aware Knowledge Graph Completion).

Type	Model	FB15K			FB15K-237			WN18			WN18RR			NELL995			DDB14		
		MRR	H@1	H@3	MRR	H@1	H@3	MRR	H@1	H@3	MRR	H@1	H@3	MRR	H@1	H@3	MRR	H@1	H@3
6*Translational/Semantic-based	TransE	96.2	94.0	98.2	96.6	94.6	98.4	97.1	95.5	98.4	78.4	66.9	87.0	84.1	78.1	88.9	96.6	94.8	98.0
	ComplEx	90.1	84.4	95.2	92.4	87.9	97.0	98.5	97.9	99.1	84.0	77.7	88.0	70.3	62.5	76.5	95.3	93.1	96.8
	DistMult	66.1	43.9	86.8	87.5	80.6	93.6	78.6	58.4	98.7	84.7	78.7	89.1	63.4	52.4	72.0	92.7	88.6	96.1
	RotatE	97.9	96.7	98.6	97.0	95.1	98.0	98.4	97.9	98.6	79.9	73.5	82.3	72.9	69.1	75.6	95.3	93.4	96.4
	SimplE	98.3	97.2	99.1	97.1	95.5	98.7	97.2	96.4	97.6	73.0	65.9	75.5	71.6	67.1	74.8	92.4	89.2	94.8
	QuatE	98.3	97.2	99.1	97.4	95.8	98.8	98.1	97.5	98.3	82.3	76.7	85.2	75.2	70.6	78.3	94.6	92.2	96.2
2*Rule-based	DRUM	94.5	94.5	97.8	95.9	90.5	95.8	96.9	95.6	98.0	85.4	77.8	91.2	71.5	64.0	74.0	95.8	93.0	98.7
7*GNN-based	R-GCN	69.6	60.1	76.0	93.1	90.3	94.2	90.9	82.4	97.9	82.2	75.0	84.8	78.5	71.2	82.6	96.2	93.8	97.9
	Path	93.7	91.8	95.1	97.2	95.7	98.6	98.1	97.1	98.9	93.3	89.7	96.1	73.7	68.5	76.4	96.9	94.8	99.1
	Con	96.2	93.4	98.8	97.8	96.1	99.5	96.0	92.7	99.2	94.3	89.4	99.3	87.5	81.5	92.8	97.7	96.1	99.4
	PathCon	98.4	97.4	99.5	97.9	96.4	99.4	99.3	98.8	99.8	97.4	95.4	99.4	89.6	84.4	94.1	98.0	96.6	99.5
	FMS	99.6	99.5	99.8	99.8	99.7	99.9	99.7	99.5	99.9	99.9	99.8	99.9	99.1	98.6	99.5	99.8	99.7	99.9

Table 10: Aggregation scores for knowledge graph models.

Model	ED	TP	VQ	BS	ZS	MO	WA	MRR	H1	H3
FMS	1.0000	1.0000	1.0000	1.0000	1.0000	0.3195	1.0000	1.0000	1.0000	1.0000
PathCon	0.8450	0.8282	0.7753	0.9120	0.8545	0.3092	0.8757	0.9710	0.9533	0.9880
Con	0.7457	0.7613	0.7014	0.7199	0.7704	0.3030	0.8023	0.9524	0.9201	0.9834
Path	0.5972	0.6067	0.3982	0.5880	0.6377	0.2925	0.6622	0.9245	0.9005	0.9419
TransE	0.5662	0.6105	0.2973	0.5000	0.5713	0.2905	0.5768	0.9182	0.8778	0.9499
QuatE	0.5369	0.5848	0.3681	0.5509	0.5134	0.2883	0.5410	0.9129	0.8878	0.9281
RotatE	0.4996	0.5544	0.3080	0.4722	0.4934	0.2853	0.5168	0.9054	0.8806	0.9174
DRUM	0.4792	0.5395	0.2701	0.4120	0.4720	0.2842	0.5000	0.9030	0.8633	0.9274
ComplEx	0.4160	0.4971	0.2949	0.4120	0.4288	0.2799	0.4810	0.8873	0.8433	0.9226
SimplE	0.4170	0.5101	0.1679	0.3773	0.3463	0.2788	0.2159	0.8857	0.8565	0.9024
R-GCN	0.2476	0.3700	0.1283	0.3426	0.2710	0.2685	0.2206	0.8537	0.7921	0.8906
DistMult	0.0001	0.1776	0.0001	0.2130	0.0010	0.2489	0.0721	0.7909	0.6742	0.8954

Table 11: Ranking of knowledge graph embedding models based on MCDM aggregator methods (1 = best).

Model	ED	TP	VQ	BS	ZS	MO	WA	MRR	H1	H3
FMS	1	1	1	1	1	1	1	1	1	1
PathCon	2	2	2	2	2	2	2	2	2	2
Con	3	3	3	3	3	3	3	3	3	3
Path	4	5	4	4	4	4	4	4	4	5
TransE	5	4	7	6	5	5	5	5	7	4
QuatE	6	6	5	5	6	6	6	6	6	6
RotatE	7	7	6	7	7	7	7	7	6	9
DRUM	8	8	9	8	8	8	8	8	8	7
ComplEx	10	10	8	8	9	9	9	9	10	8
SimplE	9	9	10	10	10	10	11	10	9	10
R-GCN	11	11	11	11	11	11	10	11	11	12
DistMult	12	12	12	12	12	12	12	12	12	11

confirm that all MCDM aggregators reliably preserve the information in the original evaluation metrics, with Z-Score and MOORA demonstrating the highest consistency.

Figure 19(a) and Table 18 present the consistency results. All aggregators achieve near-perfect normalized consistency scores (≈ 1.0) with very low variation. MOORA attains the highest overall correlation, followed closely by Z-Score and EDAS. VIKOR shows a marginal drop. This indicates stronger and more uniform preservation of base metric rankings compared to tail prediction.

For relation prediction, MOORA and Z-Score are perfectly correlated and align closely with EDAS,

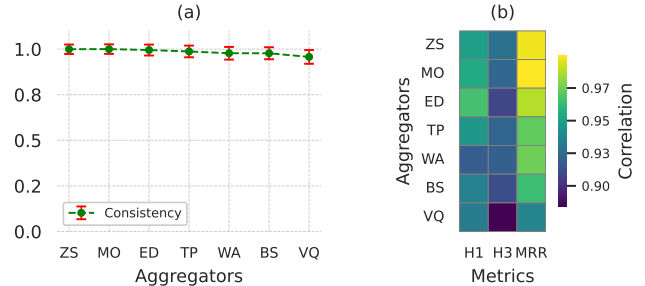


Figure 12: Relation consistency (a) Consistency test of MCDM aggregators. (b) Correlation between aggregators and base metrics.

TOPSIS, and WASPAS, indicating strong consensus among top aggregators. VIKOR remains the most divergent, consistent with its compromise-oriented ranking strategy. Overall, these results confirm that high-performing aggregators preserve rather than contradict the underlying metric signals.

Stability Test: For relation prediction, the stability test follows the same cross-dataset variance procedure described in the tail prediction stability analysis.

Figure 20 shows the distribution of ranking variance for each aggregator across the relation prediction datasets. A compact distribution indicates

Table 12: Correlation Matrices for MCDM Methods with Key Metrics, (H1, H3, MRR)

MCDM	Kendall τ ($\uparrow$)			Pearson r ($\uparrow$)			Spearman ρ ($\uparrow$)			Overall
	H1	H3	MRR	H1	H3	MRR	H1	H3	MRR	Corr.
MOORA	0.9091	0.8788	1.0000	0.9814	0.9212	0.9989	0.9720	0.9650	1.0000	0.9585
Z-Score	0.9091	0.8788	1.0000	0.9652	0.9394	0.9920	0.9720	0.9650	1.0000	0.9579
EDAS	0.9394	0.8485	0.9697	0.9783	0.9259	0.9973	0.9790	0.9510	0.9930	0.9536
TOPSIS	0.9091	0.8788	0.9394	0.9687	0.9268	0.9906	0.9580	0.9580	0.9860	0.9462
WASPAS	0.8788	0.8485	0.9697	0.9172	0.9563	0.9592	0.9580	0.9510	0.9930	0.9369
Borda	0.9313	0.8397	0.9619	0.8924	0.9459	0.9372	0.9807	0.9492	0.9912	0.9366
VIKOR	0.9394	0.7879	0.9091	0.8772	0.9556	0.9283	0.9790	0.9091	0.9720	0.9175

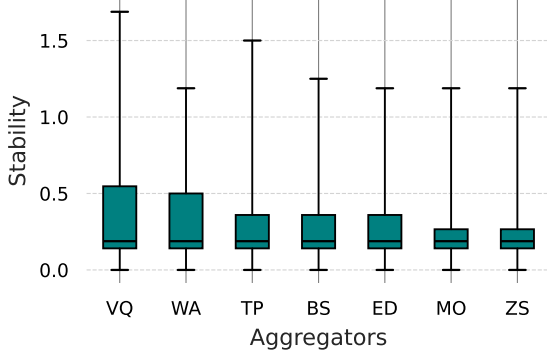


Figure 13: Distribution of ranking variance across relation prediction datasets for different aggregators.

that the aggregator produces more stable rankings across benchmarks, whereas a wider box or longer whiskers indicate greater dataset-dependent fluctuation. Z-Score and MOORA exhibit the most compact distributions, suggesting that their rankings remain more consistent across benchmark changes. In contrast, VIKOR and WASPAS show wider distributions and longer upper ranges. This suggests that their rankings are more sensitive to dataset-level changes. The main finding of stability shows that Z-Score and MOORA provide the most stable rankings for relation prediction.

Independency Test: To assess aggregator robustness to partial information loss in relation prediction, we employ a leave-one-metric-out analysis following the same procedure as for tail prediction. Each base metric is systematically omitted, and the resulting aggregator rankings are compared against the baseline computed using the complete metric set.

Figure 21 presents the normalized independency scores with error bars representing ± 1 standard deviation, sorted from highest to lowest correlation. VIKOR, WASPAS, Borda Score, and Z-Score achieve the highest independency with minimal variation, indicating strong resistance to the removal of any single metric. MOORA, EDAS, and TOPSIS fol-

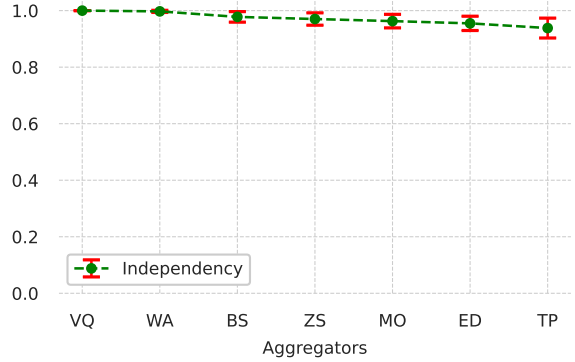


Figure 14: Metric independence of MCDM aggregators under leave-one-metric-family-out evaluation for relation prediction.

low closely with near-perfect scores, though TOPSIS shows slightly higher variability. Overall, the independency test demonstrates that relation prediction rankings remain highly stable under metric removal, with VIKOR, Borda, and Z-Score providing the strongest metric independence.

Robustness Test: The robustness test for relation prediction follows the same noise-injection procedure described in the tail prediction analysis.

Figure 21 shows the robustness scores with error bars representing ± 1 standard deviation, sorted from highest to lowest. TOPSIS and MOORA achieve near-perfect robustness with minimal variation, followed closely by EDAS and Z-Score. VIKOR shows the lowest robustness and the widest error bounds, indicating substantially higher sensitivity to noise injection. The tight error margins for top-performing methods confirm that distance- and score-based aggregators reliably preserve the ranking structure, even under substantial metric variation, for relation prediction.

For more detail, Figure 23 presents two views: the left panel shows ranks changing in an inverted order, with higher being better, and the right panel shows

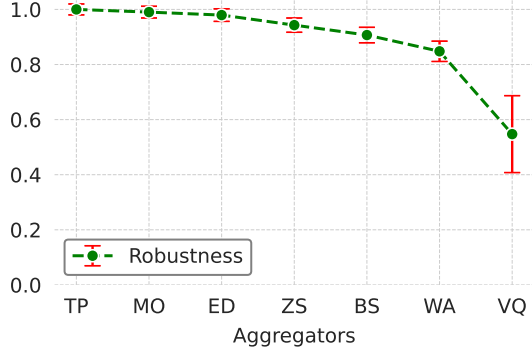


Figure 15: Robustness analysis under noise injection.

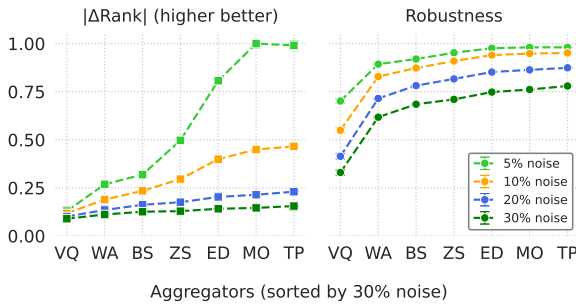


Figure 16: Robustness assessment with noise injection (1,000 iterations per level). *Left*: Mean absolute rank displacement ($|\Delta\text{Rank}|$, higher = more robust). *Right*: average correlation with baseline ranking (higher = more robust).

average correlation with the baseline ranking. TOPSIS and MOORA are the most robust aggregators, maintaining high scores across both measures as noise increases due to their ratio-based scoring, which preserves relative separation among models. In contrast, VIKOR is the most sensitive, as its compromise-ranking mechanism amplifies small metric variations. Overall, TOPSIS and MOORA preserve ranking consistency more reliably under noise, whereas VIKOR is more affected.

Generalizability Test: The relation prediction generalizability test follows the same leave-one-dataset-out validation procedure as tail prediction, examining whether an aggregator can predict rankings for an unseen benchmark using other existing results.

Figure 24 presents the cross-dataset generalizability of MCDM aggregators for relation prediction. Z-Score, TOPSIS, MOORA, Borda, EDAS, and WASPAS achieve near-perfect normalized scores (≈ 1.0) with minimal variance, demonstrating consistent ranking transferability across unseen bench-

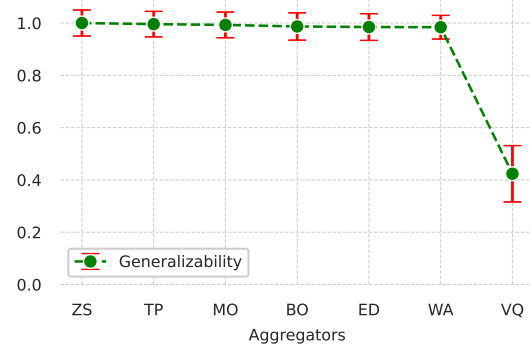


Figure 17: Generalizability analysis of MCDM aggregators under leave-one-dataset-out validation.

Table 13: Generalizability results for relation prediction using leave-one-dataset-out validation. Higher values indicate better cross-dataset transferability.

Aggregator	Top-1	Top-3	τ	Pearson	Spearman	95% CI
Mean Z-Score	1.000	1.000	0.510	0.634	0.634	[0.453, 0.568]
TOPSIS	1.000	1.000	0.517	0.635	0.635	[0.458, 0.576]
WASPAS	1.000	1.000	0.498	0.630	0.630	[0.446, 0.551]
Borda Score	1.000	1.000	0.503	0.630	0.630	[0.443, 0.564]
MOORA	1.000	1.000	0.510	0.632	0.632	[0.452, 0.568]
EDAS	1.000	1.000	0.505	0.627	0.627	[0.445, 0.565]
VIKOR	1.000	1.000	0.210	0.273	0.273	[0.112, 0.309]

marks. VIKOR exhibits a sharp decline with notably larger error bounds, indicating poor generalization and high sensitivity to dataset distribution shifts.

Table 13 reports the detailed results, where all aggregators achieve perfect Top-1 and Top-3 accuracy, with 95% confidence intervals. Overall, distance- and score-based aggregators reliably preserve model rankings across unseen benchmarks, whereas VIKOR’s compromise-based method struggles with cross-dataset transfer.

1.2.1 Pareto Analysis

Combining the five evaluation dimensions (Figure 25), Z-Score again achieves the best overall trade-off, followed by MOORA, EDAS, and TOPSIS. VIKOR remains the weakest.

Figure 25 summarizes the overall trade-off among the aggregators after combining the normalized scores from the five evaluation dimensions. Z-Score achieves the strongest overall Pareto performance, followed closely by MOORA, EDAS, and TOPSIS. This indicates that these methods maintain strong performance across multiple criteria rather than performing well in only one test. Borda Score and WASPAS remain competitive but show slightly weaker overall balance. In contrast, VIKOR shows the weakest overall Pareto performance, mainly because its strength

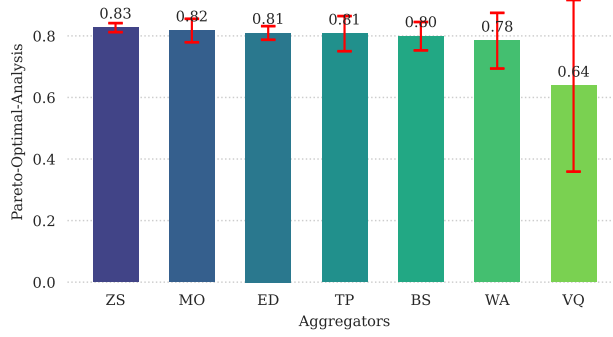


Figure 18: Pareto-based overall performance of MCDM aggregators for relation prediction.

Table 14: Top-3 KGC models ranked by the Z-Score aggregator for relation prediction (1 = best).

Model	Z-Score		
	FMS	PathCon	Con
Rank	1	2	3

in metric independence is not enough to make up for its weaker performance in robustness and generalizability. Overall, the Pareto analysis identifies Z-Score as the most suitable aggregator; it remains consistently strong across the full evaluation framework, which makes it the most balanced and reliable choice for unified relation prediction ranking.

Using the Pareto-selected Z-Score aggregator, Table 20 identifies FMS as the top-performing KGC model for relation prediction, followed by PathCon.

2 Relation Prediction: All-KGC+LOMO+LOGO analysis

2.1 Relation Prediction

We apply the same meta-evaluation pipeline (introduced in Section ?? and fully detailed in the tail prediction analysis) to the relation prediction task $(h, ?, t)$. This task uses MRR, Hits@1, and Hits@3 across six benchmark datasets: FB15K, FB15K-237, WN18, WN18RR, NELL995, and DDB14. Base results are reported in Table 15, while MCDM aggregation scores and induced rankings are presented in Tables 16 and 17.

Consistency Test: To compute consistency for relation prediction, we follow the same procedure described in the tail prediction consistency test.

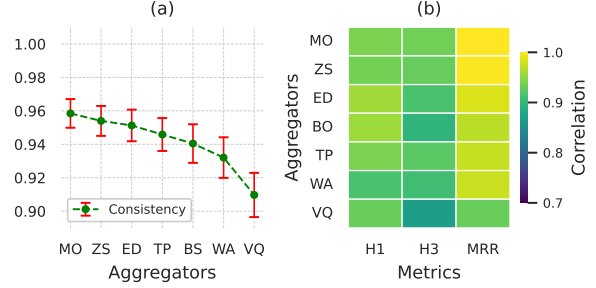


Figure 19: Relation consistency (a) Consistency test of MCDM aggregators. (b) Correlation between aggregators and base metrics.

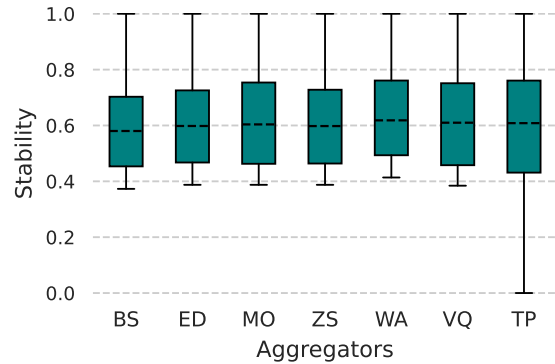


Figure 20: Distribution of ranking variance across relation prediction datasets for different aggregators.

Figure 19 (a) shows consistency scores with standard-deviation error bars. MOORA and Z-score have the highest consistency scores, with minimal variation, while VIKOR shows the lowest consistency among the others. The reason VIKOR performs worse is the calculation method. Figure 19 (b) shows a comparison between aggregators and base metrics, revealing a strong correlation among others, whereas, similarly, VIKOR produces the distinct ranking behavior. More detailed results are reported in Table 18. Overall, these results confirm that all MCDM aggregators except VIKOR reliably preserve the information in the original evaluation metrics.

Stability Test: For relation prediction, the stability test follows the same cross-dataset variance procedure described in the tail prediction stability analysis.

Figure 20 shows the distribution of ranking variance for each aggregator across the relation prediction datasets. A compact distribution indicates that the aggregator produces more stable rankings across benchmarks, whereas a wider box or longer whiskers indicate greater dataset-dependent fluctuation. Z-score and MOORA exhibit the most compact

Table 15: Transductive performance comparison on all six datasets (results are taken from Flow-Modulated Scoring for Semantic-Aware Knowledge Graph Completion).

Type	Model	FB15K			FB15K-237			WN18			WN18RR			NELL995			DDB14		
		MRR	H@1	H@3	MRR	H@1	H@3	MRR	H@1	H@3	MRR	H@1	H@3	MRR	H@1	H@3	MRR	H@1	H@3
6*Translational/Semantic-based	TransE	96.2	94.0	98.2	96.6	94.6	98.4	97.1	95.5	98.4	78.4	66.9	87.0	84.1	78.1	88.9	96.6	94.8	98.0
	ComplEx	90.1	84.4	95.2	92.4	87.9	97.0	98.5	97.9	99.1	84.0	77.7	88.0	70.3	62.5	76.5	95.3	93.1	96.8
	DistMult	66.1	43.9	86.8	87.5	80.6	93.6	78.6	58.4	98.7	84.7	78.7	89.1	63.4	52.4	72.0	92.7	88.6	96.1
	RotatE	97.9	96.7	98.6	97.0	95.1	98.0	98.4	97.9	98.6	79.9	73.5	82.3	72.9	69.1	75.6	95.3	93.4	96.4
	SimplE	98.3	97.2	99.1	97.1	95.5	98.7	97.2	96.4	97.6	73.0	65.9	75.5	71.6	67.1	74.8	92.4	89.2	94.8
	QuatE	98.3	97.2	99.1	97.4	95.8	98.8	98.1	97.5	98.3	82.3	76.7	85.2	75.2	70.6	78.3	94.6	92.2	96.2
2*Rule-based	DRUM	94.5	94.5	97.8	95.9	90.5	95.8	96.9	95.6	98.0	85.4	77.8	91.2	71.5	64.0	74.0	95.8	93.0	98.7
7*GNN-based	R-GCN	69.6	60.1	76.0	93.1	90.3	94.2	90.9	82.4	97.9	82.2	75.0	84.8	78.5	71.2	82.6	96.2	93.8	97.9
	Path	93.7	91.8	95.1	97.2	95.7	98.6	98.1	97.1	98.9	93.3	89.7	96.1	73.7	68.5	76.4	96.9	94.8	99.1
	Con	96.2	93.4	98.8	97.8	96.1	99.5	96.0	92.7	99.2	94.3	89.4	99.3	87.5	81.5	92.8	97.7	96.1	99.4
	PathCon	98.4	97.4	99.5	97.9	96.4	99.4	99.3	98.8	99.8	97.4	95.4	99.4	89.6	84.4	94.1	98.0	96.6	99.5
	FMS	99.6	99.5	99.8	99.8	99.7	99.9	99.7	99.5	99.9	99.9	99.8	99.9	99.1	98.6	99.5	99.8	99.7	99.9

Table 16: Aggregation scores for knowledge graph models.

Model	ED	TP	VQ	BS	ZS	MO	WA	MRR	H1	H3
FMS	1.0000	1.0000	1.0000	1.0000	1.0000	0.3195	1.0000	1.0000	1.0000	1.0000
PathCon	0.8450	0.8282	0.7753	0.9120	0.8545	0.3092	0.8757	0.9710	0.9533	0.9880
Con	0.7457	0.7613	0.7014	0.7199	0.7704	0.3030	0.8023	0.9524	0.9201	0.9834
Path	0.5972	0.6067	0.3982	0.5880	0.6377	0.2925	0.6622	0.9245	0.9005	0.9419
TransE	0.5662	0.6105	0.2973	0.5000	0.5713	0.2905	0.5768	0.9182	0.8778	0.9499
QuatE	0.5369	0.5848	0.3681	0.5509	0.5134	0.2883	0.5410	0.9129	0.8878	0.9281
RotatE	0.4996	0.5544	0.3080	0.4722	0.4934	0.2853	0.5168	0.9054	0.8806	0.9174
DRUM	0.4792	0.5395	0.2701	0.4120	0.4720	0.2842	0.5000	0.9030	0.8633	0.9274
ComplEx	0.4160	0.4971	0.2949	0.4120	0.4288	0.2799	0.4810	0.8873	0.8433	0.9226
SimplE	0.4170	0.5101	0.1679	0.3773	0.3463	0.2788	0.2159	0.8857	0.8565	0.9024
R-GCN	0.2476	0.3700	0.1283	0.3426	0.2710	0.2685	0.2206	0.8537	0.7921	0.8906
DistMult	0.0001	0.1776	0.0001	0.2130	0.0010	0.2489	0.0721	0.7909	0.6742	0.8954

Table 17: Ranking of knowledge graph embedding models based on MCDM aggregator methods (1 = best).

Model	ED	TP	VQ	BS	ZS	MO	WA	MRR	H1	H3
FMS	1	1	1	1	1	1	1	1	1	1
PathCon	2	2	2	2	2	2	2	2	2	2
Con	3	3	3	3	3	3	3	3	3	3
Path	4	5	4	4	4	4	4	4	4	5
TransE	5	4	7	6	5	5	5	5	7	4
QuatE	6	6	5	5	6	6	6	6	6	6
RotatE	7	7	6	7	7	7	7	7	6	9
DRUM	8	8	9	8	8	8	8	8	8	7
ComplEx	10	10	8	8	9	9	9	9	10	8
SimplE	9	9	10	10	10	10	11	10	9	10
R-GCN	11	11	11	11	11	11	10	11	11	12
DistMult	12	12	12	12	12	12	12	12	12	11

distributions, suggesting that their rankings remain more consistent across benchmark changes. In contrast, VIKOR and WASPAS show wider distributions and longer upper ranges. This suggests that their rankings are more sensitive to dataset-level changes. The main finding of stability shows that Z-score and MOORA provide the most stable rankings for relation prediction.

Independency Test: To assess aggregator robustness to partial information loss in relation prediction, we employ a leave-one-metric-out analysis following the same procedure as for tail prediction. Each base metric is systematically omitted, and the resulting aggregator rankings are compared against

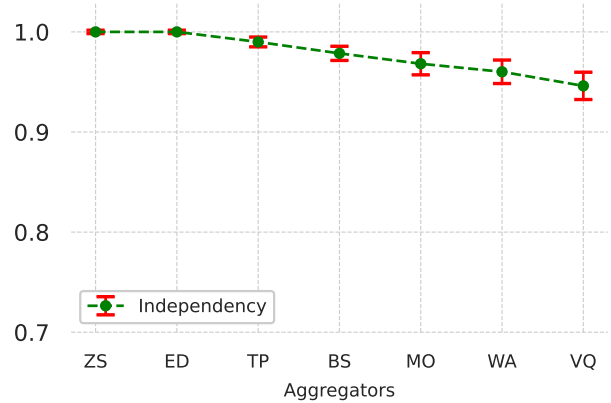


Figure 21: Metric independence of MCDM aggregators under leave-one-metric-family-out evaluation for relation prediction.

the baseline computed using the complete metric set.

Figure 21 presents independency scores with error bars representing ± 1 standard deviation, sorted from highest to lowest correlation. Z-score and EDAS achieve the highest independency followed by TOPSIS, with minimal variation, indicating strong resistance to the removal of metrics. Overall, the inde-

Table 18: Correlation Matrices for MCDM Methods with Key Metrics, (H1, H3, MRR)

MCDM	Kendall τ ($\uparrow$)			Pearson r ($\uparrow$)			Spearman ρ ($\uparrow$)			Overall
	H1	H3	MRR	H1	H3	MRR	H1	H3	MRR	Corr.
MOORA	0.8967	0.8800	1.0000	0.9625	0.9623	1.0000	0.9625	0.9623	1.0000	0.9585
Z-score	0.8908	0.8740	0.9940	0.9580	0.9578	0.9978	0.9580	0.9578	0.9978	0.9540
EDAS	0.9289	0.8478	0.9678	0.9714	0.9460	0.9910	0.9714	0.9460	0.9910	0.9513
TOPSIS	0.9084	0.8683	0.9474	0.9572	0.9511	0.9859	0.9572	0.9511	0.9859	0.9458
WASPAS	0.8566	0.8443	0.9553	0.9418	0.9400	0.9843	0.9418	0.9400	0.9843	0.9321
Borda	0.9231	0.8224	0.9428	0.9714	0.9336	0.9822	0.9730	0.9325	0.9829	0.9404
VIKOR	0.8964	0.7904	0.8895	0.9486	0.9036	0.9533	0.9486	0.9036	0.9533	0.9097

Table 19: Independency analysis of MCDM aggregators under metric removal for Relation Prediction. Kendall’s τ , Spearman’s ρ , and Pearson correlation indicate global ranking preservation (higher is better), and average absolute rank displacement ($|\Delta\text{Rank}|$) indicates local positional changes (lower is better).

Removed	Metric	Borda	Z-Mean	TOPSIS	VIKOR	EDAS	MOORA	WASPAS
3*Kendall τ	Without MRR	0.992	0.970	1.000	0.909	0.970	0.939	0.909
	Without H@1	0.992	1.000	0.970	0.879	1.000	0.939	0.909
	Without H@3	0.870	1.000	0.939	0.909	1.000	0.909	0.939
3*Spearman ρ	Without MRR	0.998	0.993	1.000	0.972	0.993	0.986	0.972
	Without H@1	0.998	1.000	0.993	0.944	1.000	0.986	0.972
	Without H@3	0.963	1.000	0.986	0.972	1.000	0.972	0.979
3*Pearson	Without MRR	0.997	0.993	1.000	0.972	0.993	0.986	0.972
	Without H@1	0.997	1.000	0.993	0.944	1.000	0.986	0.972
	Without H@3	0.954	1.000	0.986	0.972	1.000	0.972	0.979
3* $ \Delta\text{Rank} $	Without MRR	0.083	0.167	0.000	0.333	0.167	0.333	0.500
	Without H@1	0.083	0.000	0.167	0.667	0.000	0.333	0.500
	Without H@3	0.750	0.000	0.333	0.333	0.000	0.500	0.333

pendency test demonstrates that relation prediction rankings remain highly stable under metric removal, with Z-score and EDAS providing the strongest metric independence while VIKOR provides a very low independency score.

Table 19 reports the per-scenario correlation and rank displacement values. Z-Mean and EDAS demonstrate the highest robustness across all metric-removal scenarios and zero-rank displacement when H@1 or H@3 is removed. The removal of H@3 is the most disruptive overall (average $|\Delta\text{Rank}| = 0.321$), while MRR removal has the mildest impact. Overall, Z-Mean and EDAS are the most metric-independent aggregators, followed by TOPSIS and Borda, making them suitable for relation prediction tasks with incomplete or fluctuating evaluation data.

Robustness Test: The robustness test for relation prediction follows the same noise-injection procedure described in the tail prediction analysis.

Figure 21 shows the robustness scores with error bars representing ± 1 standard deviation, sorted from highest to lowest. TOPSIS and MOORA achieve strong robustness with minimal variation, followed closely by Z-score. VIKOR again shows the lowest robustness, indicating substantially higher sensitivity

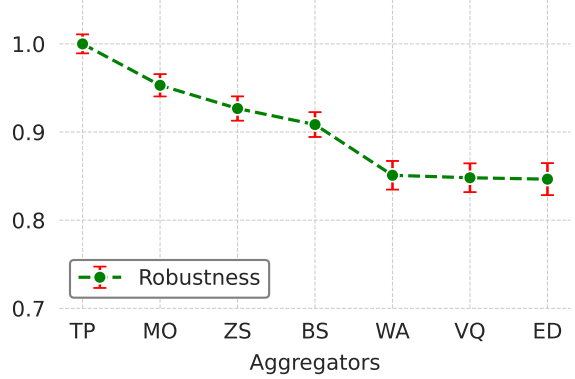


Figure 22: Robustness analysis under noise injection.

to noise injection.

For more detail, Figure 23 presents two views: the left panel shows ranks changing in an inverted order, with higher being better, and the right panel shows average correlation with the baseline ranking. TOPSIS and MOORA are the most robust aggregators, maintaining high scores across both measures as noise increases due to their ratio-based scoring, which preserves relative separation among models. In contrast, VIKOR is the most sensitive, as its compromise-ranking mechanism amplifies small metric variations. Overall, TOPSIS and MOORA preserve ranking consistency more reliably under noise, whereas VIKOR is more affected.

Generalizability Test: The relation prediction generalizability test follows the same leave-one-dataset-out validation, as tail prediction, examining whether an aggregator can predict rankings for an unseen benchmark using other existing results under KGC method removal robustness.

Figure 24 presents the cross-dataset generalizability of aggregators for relation prediction. The Z-score achieves strong generalizability with minimal variance, demonstrating consistent ranking transferability across unseen benchmarks. VIKOR exhibits poor generalizability and high sensitivity to dataset distribution shifts, among other issues.

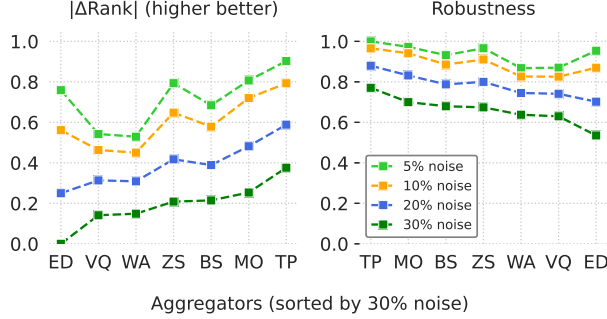


Figure 23: Robustness assessment with noise injection (1,000 iterations per level). *Left*: Mean absolute rank displacement ($|\Delta\text{Rank}|$, higher = more robust). *Right*: average correlation with baseline ranking (higher = more robust).

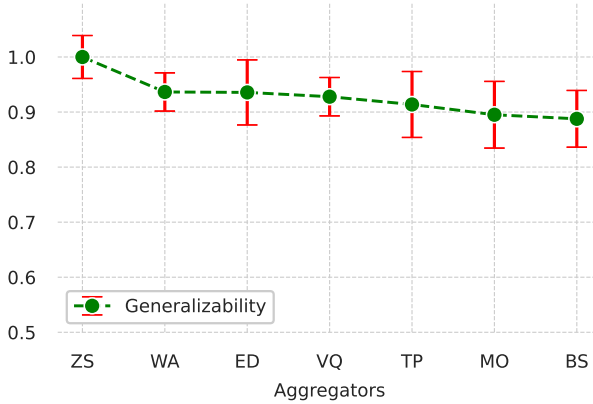


Figure 24: Generalizability analysis of MCDM aggregators under leave-one-dataset-out validation.

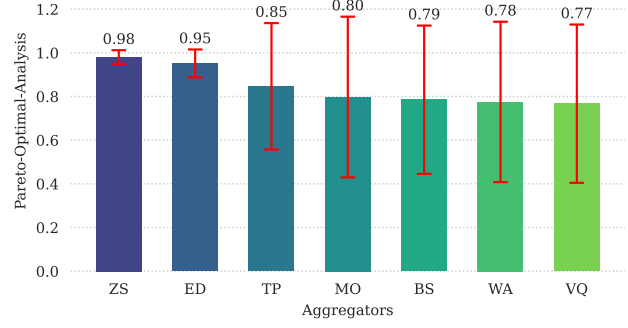


Figure 25: Pareto-based overall performance of MCDM aggregators for relation prediction.

Table 20: Top-3 KGC models ranked by the Z-score aggregator for relation prediction (1 = best).

Model	Z-score		
	FMS	PathCon	Con
Rank	1	2	3

2.1.1 Pareto Analysis

Combining the five evaluation dimensions (Figure 25), Z-score again achieves the best overall trade-off, followed by EDAS, TOPSIS, and MOORA. VIKOR remains poor.

Figure 25 summarizes the overall trade-off among the aggregators after combining the normalized scores from the five evaluation dimensions. Z-score achieves the strongest overall Pareto performance. In contrast, VIKOR shows the weakest Pareto performance. Overall, the Pareto analysis identifies Z-score as the most suitable aggregator.

Using the Pareto-selected Z-score aggregator, Table 20 identifies FMS as the top-performing KGC model for relation prediction, followed by PathCon.

2.2 Statistical Comparison (Z-Score vs MOORA)

This section provides the brief detail comparing the performance of Z-Score and MOORA aggregators across five evaluation criteria for relation and relation prediction tasks. For relation prediction (Figure 26), Z-Score achieved a higher mean performance ($M = 0.9809$) compared to MOORA ($M = 0.9650$), outperforming it on three criteria: independence, robustness, and generalizability, while tying on stability. However, statistical tests confirmed that this difference could easily happen by chance ($p = 0.273$). Even so, the moderate effect size (Cohen's $d = 0.564$)

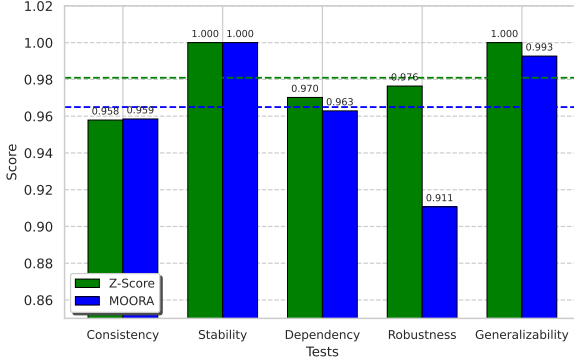


Figure 26: Comparison of Z-Score and MOORA aggregators on relation prediction. Z-Score outperforms MOORA on four of five criteria, with MOORA showing competitive results only on Stability.

shows that Z-Score still has a meaningful real-world advantage, especially in robustness.

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